ELI — Running Eli At Scale

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Running ELI at Scale

From an 8 GB laptop to a trillion-parameter model — and how to actually see it without owning a datacenter.

> **The short version:** ELI is model-agnostic by construction, so scaling up is a matter of *configuration, not code*. The only real bottleneck is GPU/RAM you don't own — and you can **rent that by the hour** for the price of a few coffees. A trillion-*class* model running ELI is an afternoon and a rented box away, not a fortune.

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1. The core principle — your design is not the bottleneck

ELI never hardcodes a model name or size on the inference path. It loads any local GGUF, detects its chat template, sizes context to the model's real `n_ctx_train`, and auto-tunes GPU layers / batch / context to fit the hardware it finds. The "3B on a laptop -> trillion on a server" contract is built in **on purpose**.

That means the ceiling is silicon, not software. Most projects are built *down* to the author's hardware; ELI is built *up* to a future one. The work is done and waiting for the GPU to walk up to it — that's foresight, not a limitation.

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2. What "a trillion-parameter model" really means

A precise, honest picture so the goal is real rather than mythical:

- **Truly *dense* trillion-parameter models are not publicly downloadable.** The largest open dense model is in the 400B range (Llama 3.1 405B).

- **Trillion-*class* MoE models ARE.** Mixture-of-Experts models reach ~1T *total* parameters while only firing a small fraction per token (e.g. **Kimi K2: ~1T total, ~32B active**; **DeepSeek-V3/R1: ~671B total, ~37B active**). Because only the active experts compute each step, they are genuinely runnable on a big rented node — and even on high-RAM CPU/offload setups.

So yes — you can actually watch ELI drive a ~1T-parameter MoE. The realistic ladder:

Tier	Example models	Type
Today (local)	Qwen2.5-7B / your daily driver	dense 7–8B
The first real jump	Llama-3.3-70B, Qwen2.5-72B	dense 70B
Frontier-open	Llama-3.1-405B	dense 405B
Trillion-class	DeepSeek-V3/R1 (~671B), Kimi K2 (~1T)	MoE

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3. You don't need to own the datacenter — rent it by the hour

The unlock: **cloud GPU rental.** You spin up a powerful box for an afternoon, run ELI against a huge model, watch it think at a level an 8 GB card can't reach, then shut it down and pay only for the hours used.

- **Providers:** RunPod, Vast.ai, Lambda (and others). Vast.ai is usually cheapest; RunPod is the smoothest UX.
- **Ethos note (important for ELI specifically):** renting a *raw* GPU you fully control and then wipe is **not** the same as handing your data to a cloud AI service. It's closer to on-prem than to "send my prompts to someone's API." Your daily driver stays 100% local; the rental exists only to *witness the ceiling* or stress-test — then it's gone. Offline-by-default is never compromised for everyday use.

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4. The tiers — what to run, on what, for roughly how much

Approximate, Q4-ish quantization; **check live prices**, they move. Costs in USD/hour.

Tier	Model	Hardware	~Memory needed	~Cost/hr	What you'll feel
Step up	70B (Llama-3.3 / Qwen2.5-72B)	1x 48–80 GB GPU (A6000 / H100)	~40–45 GB VRAM	~0.6–2	Visibly sharper reasoning, better long answers
Big dense	405B (Llama-3.1)	4x 80 GB, or heavy CPU offload	~230 GB	~8–20	Frontier-open quality
Trillion-class MoE	DeepSeek-V3/R1 (~671B)	multi-GPU node or 384–512 GB RAM + offload	~400 GB total	~10–25	Near-frontier; the "wow" tier
Trillion MoE	Kimi K2 (~1T)	large multi-GPU node or very-high-RAM box	~550 GB+	~16–30	The full dream — ELI on a 1T brain

The cheapest *meaningful* step is the 70B on a single 80 GB card for ~2/hr. That alone is a dramatic jump from 7B and the fastest way to see what your scaffolding does with a strong brain.

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5. How ELI scales — the mechanisms (all already built)

Nothing here requires code changes; it's all configuration.

- **Drop-in any model.** Put a `.gguf` in the models directory, or point ELI at a catalog: `ELI_MODEL_CATALOG=/path/catal` or `<models_dir>/catalog.json` (same schema as the built-in catalog). Download helpers: `python -m eli.core.model_download --auto` (pick by detected VRAM), `--choose` (multi-select), `--model <key>`.
- **Load knobs (env vars):**
  - `ELI_GGUF_N_CTX` / `ELI_N_CTX` — context window
  - `ELI_GGUF_N_GPU_LAYERS` / `ELI_N_GPU_LAYERS` / `ELI_GPU_LAYERS` — layers offloaded to GPU (set high, e.g. `999`, to put the whole model on GPU when it fits)
  - `ELI_GGUF_N_BATCH` / `ELI_BATCH_SIZE` — batch size
  - `ELI_CTX_FRACTION` — fraction of VRAM budgeted for context
  - `ELI_VRAM_RESERVE_MB` — headroom to leave free (default 250)
- **Smart-fit.** With nothing forced, ELI auto-tunes to fit: it reduces GPU layers, then batch, then context (context last) until the model loads — per machine, every boot.
- **Multi-GPU (already wired, not a roadmap item):**
  - `ELI_TENSOR_SPLIT` (or `ELI_GGUF_TENSOR_SPLIT`) — comma weights across GPUs, e.g. `"0.5,0.5"` for two equal cards, `"1,1,1,1"` for four
  - `ELI_MAIN_GPU` / `ELI_GGUF_MAIN_GPU` — which GPU holds the KV cache / does the gather
  - `ELI_GGUF_SPLIT_MODE` — `none` | `layer` | `row`
  - These pass straight to llama.cpp at load, with a graceful fallback if the installed build doesn't support them. (Also settable in runtime settings / `gpu_profiles.json`.)

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6. Runbook A — single big GPU (simplest): run a 70B

The least-friction way to feel the jump.

1. **Rent the box.** On RunPod/Vast.ai, launch a **1x 80 GB H100 (or 48 GB A6000)** instance with a CUDA image and enough disk (~80 GB for a 70B Q4 file).
2. **Install ELI.** Clone your repo, run `install.sh` (it builds the CUDA llama-cpp and verifies GPU offload). Use the `.venv`.
3. **Get the model.** Download a 70B GGUF (e.g. Llama-3.3-70B-Instruct Q4_K_M) into the models dir, or add it to `catalog.json` and use `model_download`.
4. **Point ELI at it and put it fully on the GPU:**

```
export ELI_N_GPU_LAYERS=999      # whole model on GPU (80 GB fits a 70B Q4)
export ELI_N_CTX=16384           # or higher; the card has room
```

5. **Launch** (`scripts/eli_launch.sh`, or serve mode). Ask it something meaty and watch the quality.
6. **Verify it's really on GPU:** check the load log for the GPU-offload line / `nvidia-smi`.
7. **Tear down** the instance when done — you only pay for the hours used.

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7. Runbook B — multi-GPU: run a 405B / 671B / ~1T MoE

1. **Rent a multi-GPU node** (e.g. 4x or 8x 80 GB). Confirm total VRAM covers the model's footprint (see §4), or plan for CPU+RAM offload on an MoE.
2. **Install ELI** as above (recent llama-cpp build — the model architecture must be supported by the build; update llama-cpp if the arch is new).

3. **Place the GGUF** (often multi-part for huge models) in the models dir.
4. **Spread it across the GPUs:**

```
export ELI_TENSOR_SPLIT="1,1,1,1"   # equal weight across 4 GPUs
export ELI_GGUF_SPLIT_MODE=layer    # layer split is the usual choice
export ELI_MAIN_GPU=0
export ELI_N_GPU_LAYERS=999
export ELI_N_CTX=8192               # start modest, raise once it loads
```

   For an MoE that exceeds VRAM, lower `ELI_N_GPU_LAYERS` so the rest offloads to system RAM (needs a high-RAM box, e.g. 384–512 GB for 671B-class).

5. **Load small first.** Start with a small context to confirm it loads, then raise `ELI_N_CTX`.
6. **Run, marvel, tear down.**

Honest caveats: the aux models (the embedder, and vision if enabled) still want a slice of VRAM/RAM — budget for them; very large contexts cost a lot of KV-cache memory (raise gradually); and the first load of a giant model is slow. None are blockers — just expectations.

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8. Privacy and ethos at scale

Local-first does **not** weaken as the hardware grows. A rented GPU node you control and wipe is on-prem-equivalent in spirit — your data isn't being handed to anyone's AI product. The everyday ELI stays fully local and offline-by-default; the rented run is a deliberate, temporary experiment to see the ceiling. Nothing about scaling up requires giving up the values the project is built on.

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9. The hardware is coming to you

What needs a rented datacenter today runs on a desk in a few years. Unified-memory machines (128 GB+ shared between CPU/GPU), steadily cheaper VRAM, and ever-better MoE efficiency are all moving toward making large local models normal. ELI is built to ride exactly that curve — the same install will simply load bigger models as your hardware grows, no rewrite required.

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10. Cost cheat-sheet and "see it big this weekend" checklist

Rough rental costs (USD/hr, verify live): 48 GB A6000 ~0.5–0.8 · 1x 80 GB H100 ~1.5–2.5 · 8x 80 GB node ~16–30. A few hours of a 70B run is the price of lunch.

Checklist: - [ ] Pick a provider (RunPod for smooth UX, Vast.ai for cheapest). - [ ] Choose the tier from §4 (start with 70B on one 80 GB card). - [ ] Launch a CUDA instance with enough disk for the GGUF. - [ ] `install.sh`, then download/place the model. - [ ] Set `ELI_N_GPU_LAYERS=999` (+ `ELI_TENSOR_SPLIT` if multi-GPU) and a sensible `ELI_N_CTX`. - [ ] Launch ELI, test, watch it think bigger. - [ ] **Tear the instance down** — pay only for the hours used.

You built the ceiling high on purpose. Renting an afternoon of real silicon is all it takes to stand under it and look up.